

ECS Project Step 5 – Create Fargate Task Definition

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon S3** bucket and an **IAM** task role were created first because the PHP application needed permission to store uploaded files in S3 at runtime. After that, a new **Amazon ECS** Fargate task definition was registered with Linux/x86_64, **awsipc** networking, 1 vCPU, and 2 GB memory. The container was configured with the image from **Amazon ECR**, port 80, and runtime environment variables for the S3 bucket name and AWS Region. A separate task execution role was also created so **Amazon ECS** could pull the image and send logs during task startup. The task definition was then created successfully and became ready for running the web application as a task or service.

Understanding Amazon ECS Task Definition

An **Amazon ECS Task Definition** is the blueprint that tells **Amazon ECS** exactly how to run a containerized application. It exists because ECS needs to know which image to use, how much CPU and memory to allocate, which network mode to use, and what permissions the app needs at runtime. In this lab, the task definition also included environment variables so the PHP app could find the **Amazon S3** bucket and Region without changing the code. The benefit is that the same image can be deployed consistently while the configuration stays outside the application.

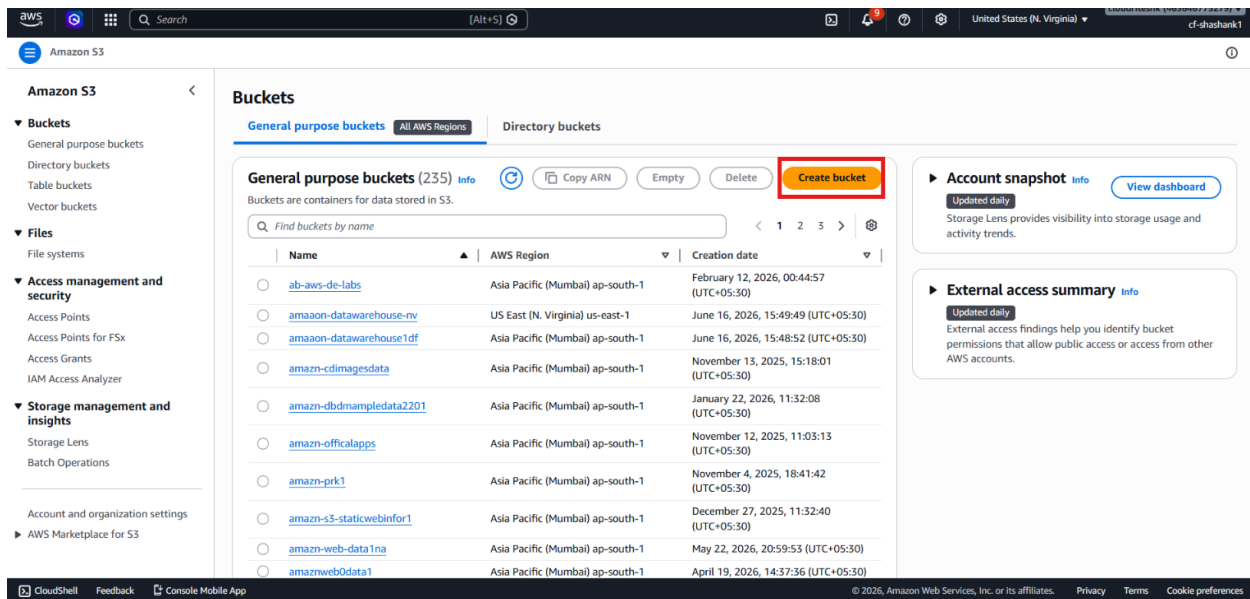
Example:

A PHP app named My-Web-App uploads output.png into a bucket named my-webapp-new. Instead of hardcoding the bucket and Region inside the code, the task definition passes `S3_BUCKET=my-webapp-new` and `AWS_REGION=ap-south-1` as environment variables. That makes the app easier to move, update, and redeploy later without rebuilding the image.

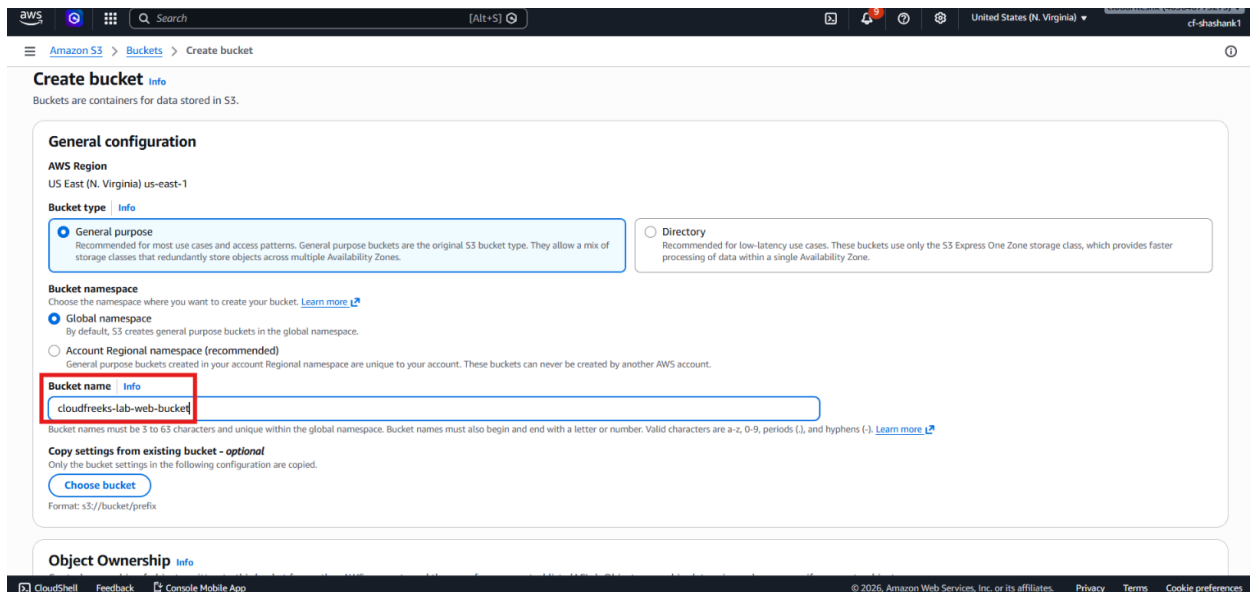
Lab Steps

Step 1 – Create the S3 bucket for application data

- Sign in to the **AWS Management Console**.
- Navigate to **Amazon S3**.
- Click **Create bucket**.

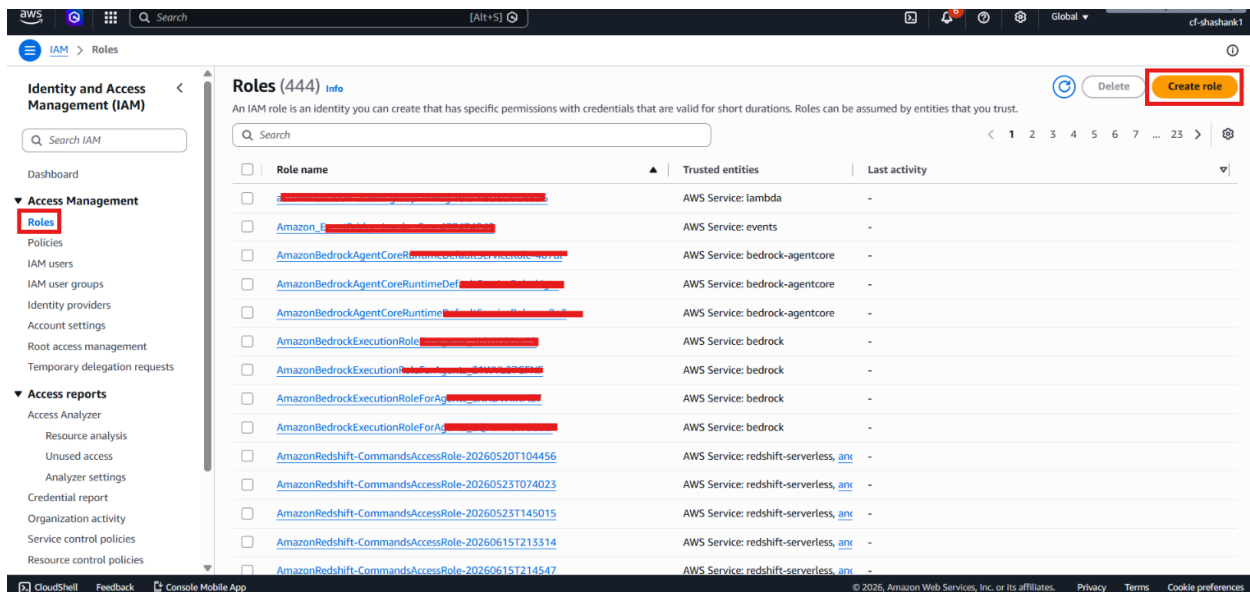


- Choose the Region where the **Amazon ECS** task will run.
- Enter a unique bucket name and note it down for later use.
- I will give name as cloudfreeks-lab-web-bucket
- Click **Create bucket**.

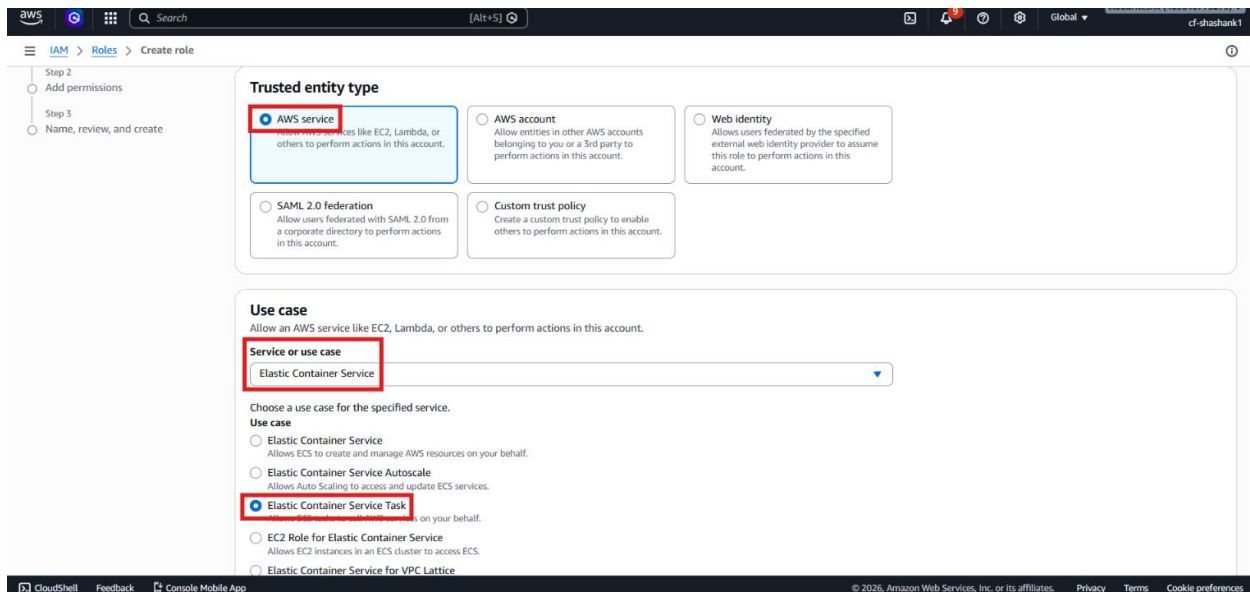


Step 2 – Create the ECS task role for S3 access

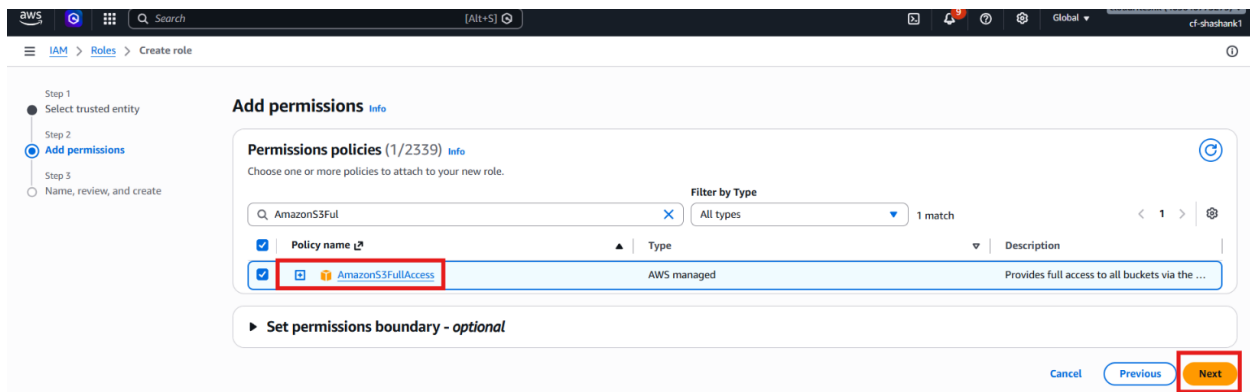
- Navigate to **IAM** → **Roles**.
- Click **Create role**.



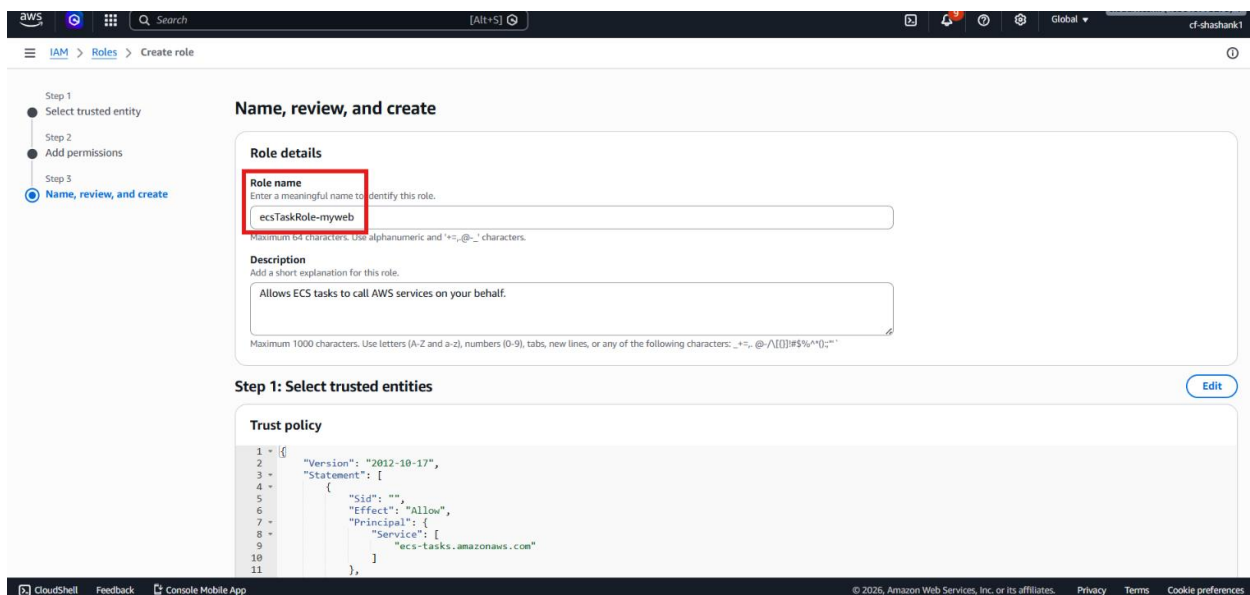
- Choose **AWS service** as the trusted entity type.
- Choose **Elastic Container Service**.
- Select **Elastic Container Service Task** as the use case.



- Attach **AmazonS3FullAccess** for testing.

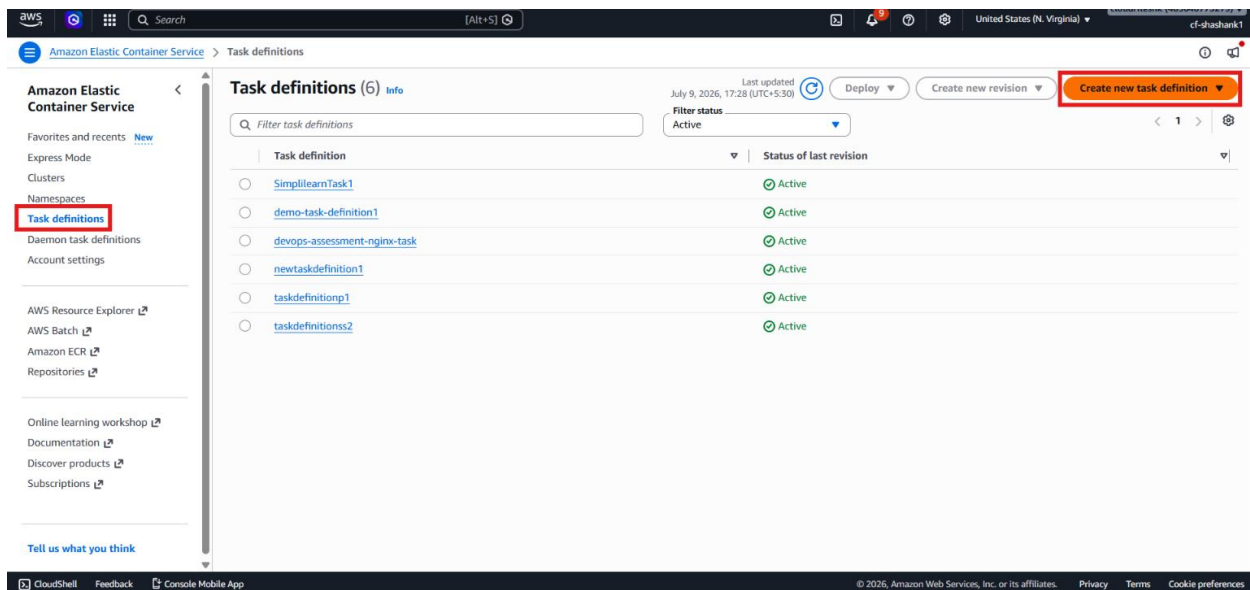


- Name the role `ecsTaskRole-myweb`.
- Click **Create role**.

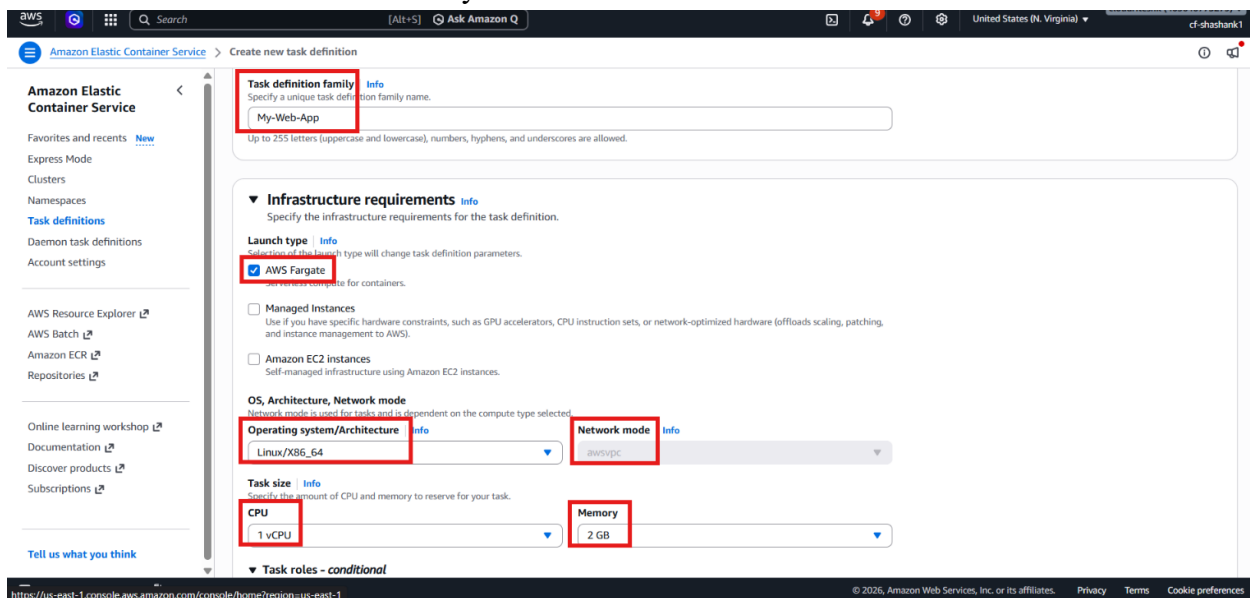


Step 3 – Create the Fargate task definition

- Open Amazon ECS → Task definitions.
- Click **Create new task definition**.



- Enter the family name: My-Web-App.
- Select **FARGATE** as the launch type compatibility.
- Choose Linux/x86_64.
- Keep the network mode as **awsvpc**.
- Set CPU to 1 vCPU and memory to 2 GB.



- Select the task role **ecsTaskRole-myweb**.
- For the task execution role, choose **Create new role**.
- Add a container named **web**.
- Paste the image URI of from **Amazon ECR** for the repo we have created earlier.

Amazon Elastic Container Service > Create new task definition

Task roles - conditional

Task role Info
A task IAM role allows containers in the task to make API requests to AWS services.
ecsTaskRole-myweb Create new role

Task execution role Info
A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM role created, we can create one for you.
Create default role Create new role

Task placement - optional

Fault injection - optional

Fault injection
☐ Enable fault injection

Container - 1 Info Essential container Remove

Container details
Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name
web Essential container: Yes

Image URI
dkr.ecr.us-east-1.amazonaws.com/cloudfreeks-s3-php Browse ECR images

- Set the container port to 80.

Amazon Elastic Container Service > Create new task definition

Image URI
dkr.ecr.us-east-1.amazonaws.com/cloudfreeks-s3-php Browse ECR images

Private registry Info
Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.
☒ Private registry authentication

Port mappings Info
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port 80 **Protocol** TCP **Port name** container-port-protocol **App protocol** HTTP Remove

Add port mapping

Read only root file system Info
When this parameter is turned on, the container is given read-only access to its root file system.
☐ Read only

Resource allocation limits - conditional Info
Container-level CPU, GPU, and memory limits are different from task-level values. They define how much resources are allocated for the container. If container attempts to exceed the memory specified in hard limit, the container is terminated.

CPU 1 **Memory hard limit** 3 **Memory soft limit** 1

Neuron device - new
Managed Neuron devices are not available with AWS Fargate launch type.

All GPUs - new
☐ Allocate all instance GPUs
GPUs are not available with AWS Fargate launch type.

GPU 1
GPUs are not available with AWS Fargate launch type.

- Add the environment variables S3_BUCKET and AWS_REGION and provide their respective values.

Amazon Elastic Container Service > Create new task definition

Neuron device - new
Managed Neuron devices are not available with AWS Fargate launch type.

All GPUs - new
Allocate all instance GPUs
GPUs are not available with AWS Fargate launch type.

GPU
GPUs are not available with AWS Fargate launch type.

Environment variables - optional
Environment variables | Info

[Bulk edit](#)

Add individually
Add environment variables using plain text values or secrets from AWS Secrets Manager or Parameter Store.

Key	Value type	Value	
S3_BUCKET	Value	cloudfrecks-lab-web-bucket	Remove
AWS_REGION	Value	us-east-1	Remove

[Add environment variable](#)

Add from file
Add environment variables in bulk by providing an environment file hosted on Amazon S3.
[Add environment file](#)
You can add 10 more environment files.

Logging - optional
1 CPU and memory allocation for a sidecar
There are logging options that will automatically add a sidecar to your task definition if it does not already exist. AWS provides CPU and memory adjustment recommendations based on the state of existing...

- Keep the default ephemeral storage setting.
- Click **Create**.

Amazon Elastic Container Service > Create new task definition

Storage - optional
Ephemeral storage | Info
The amount of ephemeral storage, in GiB, to allocate for the task. By default, your tasks hosted on AWS Fargate receive a minimum of 20 GiB of ephemeral storage.
To specify a custom amount of ephemeral storage, specify a value between 21 GiB up to a maximum of 200 GiB.

Amount
21

Volumes - Info
Add one or more data volumes for your task to provide additional storage for the containers in the task. For each data volume, you must add a mount point to specify where to mount the data volume in the container.
[Add volume](#)

Volumes from - Info
Mount data volumes from another container.
[Add volume from](#)

Monitoring - optional
Configure your application trace and metric collection settings using the AWS Distro for OpenTelemetry integration.

Tags - optional Info
Tags help you to identify and organize your task definitions.

[Cancel](#) [Create](#)

Step 4 – Confirm the task definition setup

- Verify that the task definition was registered successfully.

The screenshot displays the AWS Management Console for Amazon Elastic Container Service (ECS). The breadcrumb navigation shows the path: Amazon Elastic Container Service > Task definitions > My-Web-App > Revision 1 > Containers. A green notification banner at the top indicates that the task definition 'My-Web-App:1' has been successfully created. The main content area shows the details for 'My-Web-App:1', which is in an 'ACTIVE' state. Key details include: ARN (redacted), Task role (ecsTaskRole-myweb), Task execution role (ecsTaskExecutionRole), Time created (July 9, 2026, 17:48 UTC+5:30), App environment (Fargate), and Network mode (awsvpc). The 'Task size' section shows a Task CPU of 1,024 units (1 vCPU) and Task memory of 2,048 MiB (2 GiB). Below this, there are two horizontal bars representing the maximum allocation for CPU and memory. The CPU bar is at 1,024 units, and the memory bar is at 2,000 MiB. The left sidebar contains navigation links for Favorites and recents, Express Mode, Clusters, Namespaces, Task definitions, Daemon task definitions, Account settings, AWS Resource Explorer, AWS Batch, Amazon ECR, Repositories, Online learning workshop, Documentation, Discover products, and Subscriptions. The bottom of the console shows the CloudShell, Feedback, and Console Mobile App links, along with the copyright notice for Amazon Web Services, Inc. or its affiliates.

- Confirm that the **Amazon S3** bucket name and Region were recorded for later use.
- Confirm that the task role and task execution role were both configured correctly.
- Confirm that the container is ready to run with the **Amazon ECR** image and runtime environment variables.

Verification

- Verified that the **Amazon S3** bucket was created successfully.
- Verified that the **IAM** task role ecsTaskRole-myweb was created successfully.
- Verified that the **Amazon ECS** Fargate task definition was registered successfully.
- Verified that the task definition used **awsvpc** networking and Linux/x86_64.
- Verified that the container image was linked from **Amazon ECR**.
- Verified that the environment variables for bucket name and Region were added correctly.
- Verified that the task definition was ready to run as an ECS task or service.